

MIE 1621 Computational Project Version A Winter 2025

Instructions: Use MATLAB to do the following assignment. Submit report and all code & data through quercus by 11:49PM on April 7th (this is the official deadline) but I would allow an extension without penalty to April 11th by 11:59PM.

Part 1: Consider the following financial optimization model:

$$\begin{aligned} & \text{maximize} \quad \sum_{i=1}^n \mu_i x_i - \frac{\delta}{2} \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} x_i x_j \\ & \quad \text{subject to} \\ & \quad \sum_{i=1}^n x_i = 1 \\ & \quad x_i \geq 0, \quad i = 1, \dots, n \end{aligned}$$

We assume there are n financial assets (e.g. stocks) such that future prices (returns) of each asset is random and so each asset i has associated with it an expected return μ_i and a variance of returns σ_i^2 . Also, asset returns are pairwise correlated as embodied in the covariance quantities σ_{ij} for $i \neq j$. This model seeks to determine a portfolio of asset weights x_i (where x_i is the proportion of investment in asset i) such that the portfolio maximizes risk-adjusted expected return (the second term is a measure of risk of a portfolio x). The quantity $\delta > 0$ is a parameter that controls risk aversion (degree of risk that is tolerable for an investor). The larger δ is the more risk averse an investor is. The first constraint of the model ensures that the x_i variables are proportions and the non-negativity constraints on the variables x_i ensure that asset can only be bought not sold.

Data for the Model

Asset	1	2	3
Expected return	$\mu_1 = 10.73\%$	$\mu_2 = 7.37\%$	$\mu_3 = 6.27\%$

σ_{ij}	1	2	3
1	0.02778	0.00387	0.00021
2	0.00387	0.01112	-0.00020
3	0.00021	-0.00020	0.00115

Choose δ to be between 3.5 and 4.5.

Using the data above for the three assets implement a **predictor-corrector primal-dual interior point method** to solve this financial model. Compare your results with those obtained from using the MATLAB function quadprog see <https://www.mathworks.com/help/optim/ug/quadprog.html>

The implementation must be based on the algorithm from the lecture slides and you must NOT use any code from any other source. In your write up you have to **write out the complete primal-dual interior point algorithm step by step for the quadratic programming case in detail**. Also, you must **comment** your MATLAB code so that it is easy for us to detect that you followed the primal-dual algorithm from lecture.

Part 2 Large-scale strictly convex quadratic programming

Using your primal-dual code from Part 1 solve larger versions of the model from Part 1 EXCEPT that you will be **randomly generating the covariance matrix**. It is easy to generate a positive definite matrix in MATLAB. Here is sample MATLAB code:

```
n = 3; % Size of the matrix
A = randn(n);
A = A + A';
A = A + n*eye(n);
```

Choose **at least 5 different sizes of n from smaller to larger with n at least 3 and for each size of n create the model instance using the randomly generated covariance matrix of size n (also set $\mu_i = 1$ for $i = 1, \dots, n$)** use the same δ value used in Part 1 and solve each model (1) using your primal-dual code and (2) quadprog. Compare the differences. The larger size n you can solve well using your primal-dual code the higher your project score for Part 2.

Deliverable: You must write up a brief report which details your implementation of the predictor-corrector primal-dual interior point method as well as providing the computational results. **Show the iterates of the primal problem and the dual problem in tables for Part 1 only**. Do NOT just dump out the results from executing the code. Also, compare your results with the results from using MATLAB quadprog. You must provide the actual code in the report. The code you provide must be commented and easy to follow. **You MUST make sure that your code is easy to execute so that results can be validated. You MUST provide brief instructions on how to execute your code for the various parts of the project. If it is not easy to get your code running than marks will be lost!** Make sure that your report is well written, concise and to the point. We do not want too many pages in a report including the appendix.